American Computer Science League

2020-2021 • Contest 4: Shorts • Elementary Division

1. Graph Theory

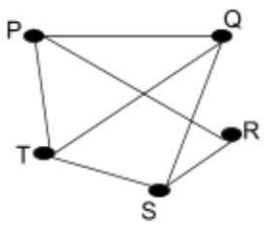
Draw an undirected graph representing the following information using the first letter of each person's name. Which person has the most friends?

Ann is friends with Deb and Eve. Eve is friends with Ann and Carol. Deb is friends with Carol and Fran. Fran is friends with Ann and Carol. Carol is friends with Beth and Ann. Beth is friends with Eve and Deb.

A. Ann
B. Beth
C. Carol
D. Deb
E. Eve

2. Graph Theory

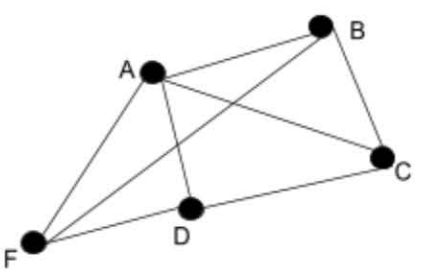
For the given graph, how many more edges are needed for this graph to be a complete graph?



A. 1
B. 2
C. 3
D. 4
E. 5

3. Graph Theory

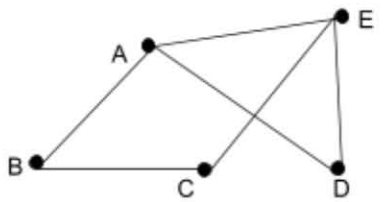
In creating a transportation network among the cities of Austin, Boston, Concord, Denver, and Frankford that uses the first letter of each city's name, how many different routes (or simple paths) are there from Austin to Denver?



A. 4
B. 5
C. 6
D. 7
E. 8

4. Graph Theory

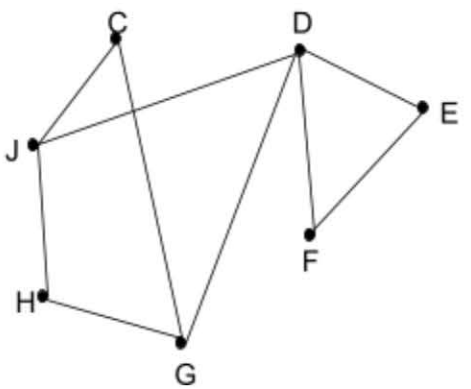
In the undirected graph shown below that represents Al's, Ben's, Cy's, Dave's, and Eli's houses using the first letter of each name, how many cycles are there?



A. 4
B. 6
C. 8
D. 10
E. 12

5. Graph Theory

Identify the starting and ending vertices that can be used for the following graph to be traversable.



A. GJ
B. DG
C. JD
D. Any of them
E. None of them

American Computer Science League

2020-2021 ● Contest 4: Shorts ● Elementary Division

1. Graph Theory

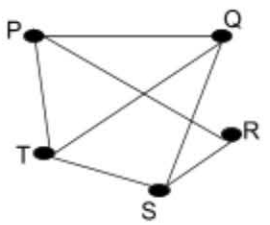
使用每个人姓名的首字母绘制一个无向图，表示以下信息。请问哪个人的朋友最多？

Ann是Deb和Eve的朋友。Eve是Ann和Carol的朋友。Deb是Carol和Fran的朋友。Fran是Ann和Carol的朋友。Carol是Beth和Ann的朋友。Beth是Eve和Deb的朋友。

A. Ann
B. Beth
C. Carol
D. Deb
E. Eve

2. Graph Theory

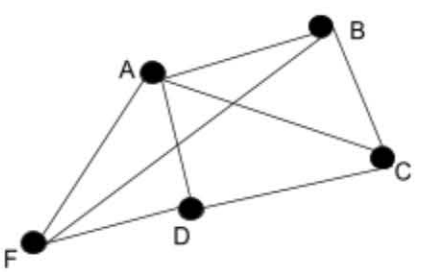
要增加多少条边才能把下面的图形变成完全图？



A. 1
B. 2
C. 3
D. 4
E. 5

3. Graph Theory

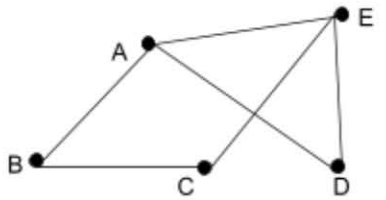
使用城市名称的首字母构建 Austin、Boston、Concord、Denver 和 Frankford 的城市交通网络，请问从 Austin 到 Denver 有多少条不同的路径（或简单路径）？



A. 4
B. 5
C. 6
D. 7
E. 8

4. Graph Theory

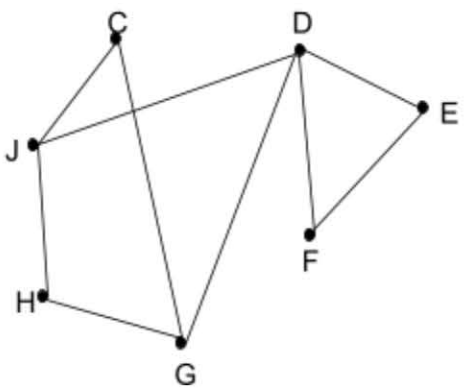
下方无向图使用姓名首字母代表Al、Ben、Cy、Dave 和 Eli 的家，请问共有多少个环？



A. 4
B. 6
C. 8
D. 10
E. 12

5. Graph Theory

找出可以遍历下图的起始顶点和结束顶点。



A. GJ
B. DG
C. JD
D. 以上都可以
E. 以上都不可以